FINDING A PARAMETER SEQUENCE WHOSE POWERS ENTER I AND WHOSE ANNIHILATORS ARE CONTROLLED

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Ogoma's annihilator lemma. lemma-ogoma (carded L26814). Produces the multiplier s in S realizing the annihilator equality at each induction step.
- Regular sequences in regular local rings. Algebra, lemma regular-ring-CM and lemma regular-sequence-powers (cross-chapter, named). Give that the parameters form a regular sequence and that powers preserve regularity.

Source reference. Original reference 00NQ: focused-seeds/geometry/S9-smo-find-sequence/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07DV: focused-seeds/geometry/S9-smo-find-sequence/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FC: sources/stacks-smoothing/pinned-original-source.tex, source lines 2504–2504. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let $I \subset \mathfrak{q}$ be a prime. Let n, e be positive integers Assume that $\mathfrak{q}^n\Lambda_\mathfrak{q} \subset I\Lambda_\mathfrak{q}$ and that $\Lambda_\mathfrak{q}$ is a regular local ring of dimension d. Then there exists an $n > 0$ and $\pi_1, \ldots, \pi_d \in \Lambda$ such that*

(1) $(\pi_1, \ldots, \pi_d)\Lambda_\mathfrak{q} = \mathfrak{q}\Lambda_\mathfrak{q}$,
(2) $\pi_1^n, \ldots, \pi_d^n \in I$, *and*
(3) *for $i = 1, \ldots, d$ we have*

$$Ann_{\Lambda/(\pi_1^e, \ldots, \pi_{i-1}^e)\Lambda}(\pi_i) = Ann_{\Lambda/(\pi_1^e, \ldots, \pi_{i-1}^e)\Lambda}(\pi_i^2).$$

Proof. Set $S = \Lambda \setminus \mathfrak{q}$ so that $\Lambda_\mathfrak{q} = S^{-1}\Lambda$. First pick $\pi_1, \ldots, \pi_d$ with (1) which is possible as $\Lambda_\mathfrak{q}$ is regular. By assumption $\pi_i^n \in I\Lambda_\mathfrak{q}$. Thus we can find $s_1, \ldots, s_d \in S$ such that $s_i\pi_i^n \in I$. Replacing π_i by $s_i\pi_i$ we get (2). Note that (1) and (2) are preserved by further multiplying by elements of S. Suppose that (3) holds for

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$i = 1, \ldots, t$ for some $t \in \{0, \ldots, d\}$. Note that $\pi_1, \ldots, \pi_d$ is a regular sequence in $S^{-1}\Lambda$, see Algebra, Lemma 00NQ. In particular $\pi_1^e, \ldots, \pi_t^e, \pi_{t+1}$ is a regular sequence in $S^{-1}\Lambda = \Lambda_{\mathfrak{q}}$ by Algebra, Lemma 07DV. Hence we see that

$$\mathrm{Ann}_{S^{-1}\Lambda/(\pi_1^e, \ldots, \pi_{i-1}^e)}(\pi_i) = \mathrm{Ann}_{S^{-1}\Lambda/(\pi_1^e, \ldots, \pi_{i-1}^e)}(\pi_i^2).$$

Thus we get (3) for $i = t+1$ after replacing π_{t+1} by $s\pi_{t+1}$ for some $s \in S$ by Lemma 07FC. By induction on t this produces a sequence satisfying (1), (2), and (3). $\square$